

**BỘ GIÁO DỤC VÀ ĐÀO TẠO
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ ĐÔNG Á**



TIỂU LUẬN

HỌC PHẦN: Management Information Systems

**ĐỀ TÀI: Optimizing User Experience in Music
Cataloging: A Study of MML - My Music List System.**

Giảng viên hướng dẫn:	MSc. Nguyễn Viết Hùng
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Lớp:	DC.BM.13.10
Mã đề:	03

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Cán bộ chấm thi 1

(Ký và ghi rõ họ tên)

Cán bộ chấm thi 2

(Ký và ghi rõ họ tên)

Abstract

The My Music List (MML) project is a comprehensive platform designed to facilitate music enthusiasts in tracking, rating, and reviewing their favorite songs. Inspired by the popular My Anime List, MML enables users to create personalized music lists, rate songs on a scale of 1 to 10, and view statistics on their musical preferences. Key features include secure user authentication, detailed song ratings, and analytics that highlight the percentage of songs rated, as well as the highest and lowest rated tracks. Utilizing modern web technologies, MML aims to provide a user-friendly interface and a robust database infrastructure to enhance user experience. This project not only serves as a personal music management tool but also fosters a community of shared musical interests, paving the way for future enhancements such as song recommendations and social sharing capabilities.

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1. Introduction

My Music List (MML) is a music management and rating app designed to help users keep track of and review their favorite songs, albums, and artists. It's inspired by My Anime List (MAL)[1], a popular platform for anime fans. While MAL has built a strong community around anime, MML is all about music lovers, offering a fresh space where users can share and discover music based on their tastes.

Unlike popular music services like Spotify and Apple Music, which have huge libraries but lack detailed rating features, MML fills that gap. It allows users to not only track their song lists but also rate and review each song in depth. It's more than just a place to save your favorite tunes—it's a growing community where music fans can connect, interact, and explore new tracks based on honest reviews from fellow users.[2]

2. Overview

MML is a user-friendly platform that lets people manage, rate, and share music in an organized way. Users can create lists of songs, albums, and artists they follow, and rate these based on their own preferences and feelings. MML also pulls in community ratings to give users a broader view of how music is being rated across the platform.

One of the standout features of MML is its ability to provide detailed stats on rating distribution. This includes the highest and lowest ratings for each song, how scores are spread, and how these trends change over time. It offers deeper insights into both personal music tastes and what the community thinks.

2.1 Vision, Mission, and Core Values

2.1.1 Vision

MML aims to become the go-to platform for personalized music management and ratings. We want to create a space where music fans can easily discover, track, and share their favorite songs, albums, and artists. Our vision is to build a vibrant music community where users not only enjoy music but also connect with others who share their love for it.

We see MML as more than just a personal music tool—it's a platform that helps people find new music and encourages sharing and reviewing based on personal experiences. We want to build an active community where people can explore the latest music trends and contribute to a diverse and rich music culture.[3]

2.1.2 Mission

Our mission at MML is to offer music lovers a comprehensive and easy-to-use platform for managing, rating, and sharing music in a personalized way. We're dedicated to improving and adding tools that help users discover new music, give high-quality reviews, and make the most of their daily music experience.

With smart analysis and recommendations based on user preferences, we aim to help people find songs that fit their moods, activities, or specific vibes. MML isn't just a music management app it's a bridge that helps users explore and connect with music all over the world.

2.1.3 Core Values

Personalization: We believe music is a personal part of life, so MML focuses on delivering a unique experience for every user. The app is designed to reflect each person's music tastes and style, making every account a true reflection of their personality.

Community: MML is not just a personal tool; it's a dynamic community where music lovers can interact, share, and learn about each other's favorite songs and artists. Connecting music fans is a core part of MML's growth.

Creativity and Discovery: We encourage users to explore new music styles and share their thoughts through reviews and comments. Discovery is one of the biggest motivations for enhancing the music experience.

Transparency and Quality: MML is committed to providing a transparent rating system where users can trust the reviews from real community experiences. We focus on quality in everything—from features to overall user experience.[4]

3. Requirements Analysis

Our system is designed with advanced features to optimize user experience and ensure data security, protecting customer info from leaks and other risks.

3.1 Login and Logout:

- **Password Encryption:** We use encryption to keep user passwords safe and prevent leaks. With online threats and hacking increasing, it's crucial to secure personal info. Encrypted passwords are harder to crack, even if exposed, making the system more secure and trustworthy.

- **Why It Matters:** In today's world, where personal data can be targeted by hackers for various scams, protecting customer information is our top priority. Encryption helps prevent unauthorized access and keeps your data safe.[5]

3.2 Song Rating

Flexible Rating: Users can rate songs from 1 to 10 based on their personal preferences, without any rules or restrictions. This flexibility makes rating easier and more personal, leading to more accurate data collection.

Data Collection: By gathering these ratings, we build a comprehensive database that helps us generate detailed, objective reviews based on user feedback. It allows us to classify songs according to user preferences and present them accordingly.

Automatic Calculation: The system automatically sorts and displays songs from highest to lowest ratings based on user feedback. This feature helps users easily find popular tracks or those less favored, enhancing their music discovery experience.[6]

3.3 Statistics

Rating Stats: The system provides stats on the number of ratings, showing the percentage of songs rated high or low. This feature helps users track their rating activity and encourages them to interact more with the system.[7]

User Engagement: By tracking and displaying these stats, users are motivated to engage more with the platform, making their experience more dynamic and connected to their music preferences.

4. Database Design

4.1 ER Model

Main Entities

Figure 1.1 represents a simple relational database schema for the "My Music List" (MML) project. It consists of three tables: **User**, **Review**, and **Song**, connected through foreign keys to form relationships between users, their reviews, and songs.

User

- UserID: Primary key, unique identifier for each user.
- Username: User's login name.
- Password: Hidden password.

Song

- SongID: Primary key, unique identifier for each song.

- Title: Song name.
- Artist: Artist name.
- Genre: Song genre.

Rating

- RatingID: Primary key, unique identifier for each rating.
- UserID: Foreign key linking to the User entity.
- SongID: Foreign key linking to the Song entity.
- Score: Rating score (1-10).



Figure 1.1 ER Model of MAL

4.2 Relationships

User - Rating: A user can rate many songs, but each rating belongs to only one user. This is a one-to-many relationship from User to Rating.

Song - Rating: A song can receive many ratings from different users, but each rating is related to only one song. This is a one-to-many relationship from Song to Rating.

4.3 Keys

Primary Keys:

- User ID for the User entity.
- RatingID: Primary key, unique identifier for each rating.
- Song ID for the Song entity.

Foreign Keys:

- User ID in the Review table links to the User table.
- Song ID in the Review table links to the Song table.

4.4 Cardinality

The arrows in the diagram make it clear that a single user can write as many reviews as they want, but each review is tied to just that one user. It's a one-to-many relationship one user, many reviews. The same thing applies to songs: one song can

have tons of reviews from different users, but each review is only linked to one specific song.

Basically, a user can go through the platform and review all the songs they love (or hate), and the system keeps track of who reviewed what. At the same time, each song could have dozens or even hundreds of reviews from different users, helping to create a broader perspective on how people feel about that song.

4.5 Data Flow

When a user logs into MML, their account is verified by checking their User ID and Password—pretty straightforward. Once logged in, they're free to browse the song library and leave reviews on the tracks they've listened to. When a review is submitted, MML records which user left the review and which song it's for by linking the User ID and Song ID. The user's rating is then stored in the Review table.

In short, every time a user leaves a review, the system automatically ties that review to their account and the song they're rating. Over time, this builds a detailed collection of user ratings, allowing other users to explore the community's opinions on different songs.[8]

5. Implementation

5.1 Technologies Used: Database, programming languages, framework

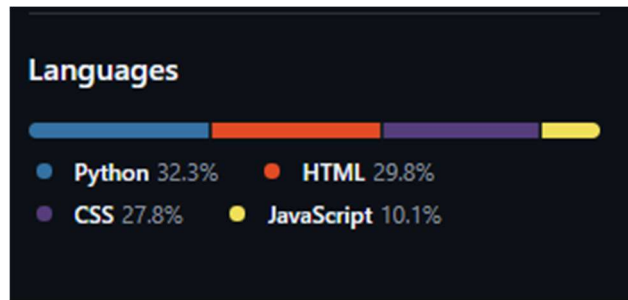
Figure 2.1 represents a breakdown of the programming languages used in a project, with corresponding percentages.

For the MML project, we kept things simple but effective when it came to technology. Here's a quick rundown of what we used:

Database: We went with SQLite for the database. It's lightweight, easy to set up, and perfect for a project like this where we're handling song ratings and user info without needing a super complex database system.

Programming Language: The backend is built using Python, specifically with the Flask framework. Flask is great because it's lightweight but still powerful enough to handle all the server-side tasks we need for MML, like handling user logins, storing song reviews, and managing the data flow.[9]

Frontend: For the frontend, we kept it clean and simple with HTML and CSS for the basic structure and design. But to make the site a bit more interactive, we added some JavaScript for things like dynamic loading, real-time updates, and making sure the user experience is smooth and responsive.[11]



(Source: github.com/Xernnn/MyMusicList)

Figure 2.1 Languages of MAL Project

5.2 Development Process: Key steps from design to implementation

Stage 1: Concept and Planning:

We started by outlining the core features of MML, like the ability for users to rate and review songs, create personalized song lists, and explore what others are listening to. We also took inspiration from platforms like My Anime List to ensure we were hitting the right notes for our target audience music lovers.

Stage 2: Designing the Look and Layout:

After that, we sketched out how the site should look and feel. The goal was to keep the design simple and user-friendly, so people could easily browse, rate songs, and check out reviews. We also worked on how the database should be structured to handle user info, song details, and reviews smoothly.

Stage 3: Outsource

We do not have enough ability to code so we decide to rent a graduation IT from USTH to develop MML based on our idea.

Stage 3.1 Building the Backend:

Once we had the design sorted, we started coding the backend using Python and Flask. This involved setting up everything behind the scenes, like user logins, song reviews, and making sure all the data was saved properly in the SQLite database.

Stage 3.2 Building the Frontend:

With the backend done, we shifted to the frontend using HTML, CSS, and some JavaScript to make things interactive. We focused on making sure the interface looked good and that everything worked, like submitting reviews, searching for songs, and navigating the site.

Stage 3.3 Putting It All Together and Testing:

After both the backend and frontend were ready, we connected them and made sure they played nice together. Then, we tested everything from basic features to user interactions to make sure there were no bugs or issues that would mess up the experience.

Stage 4: Testing

After integration, we moved on to testing. We ran unit tests to check individual components (like the review system), and integration tests to ensure everything worked well together. We also had real users try out the platform to identify any bugs or areas that needed improvement. This stage was crucial for fine-tuning both performance and the overall user experience.

6. Website Function

Users: Anyone who needs to access the information and services provided by the website must have the ability to log in and log out. These are crucial elements for securing and personalizing the user experience on online platforms. They not only help protect personal information but also facilitate interaction with the system.

Benefits of Logging into the Website:

Information Security: Logging in ensures that only registered users have access to personal information.

Account Management: Users can track various details related to their accounts.

Main Functions:

- Login:

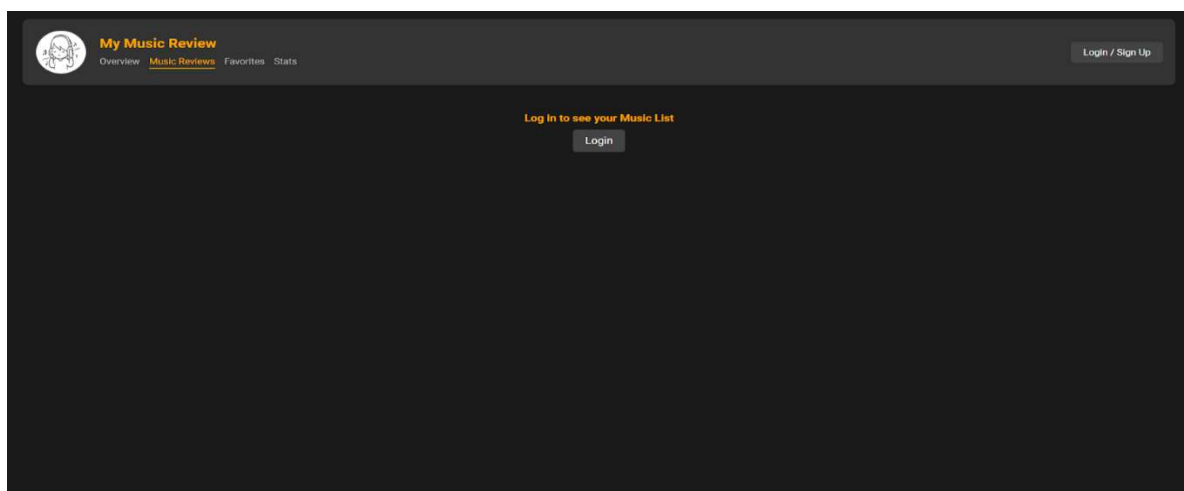


Figure 3.1 Login Interface

When users go to the "My Music Review" website, a clear message will appear on their main screen saying "Log in to see your Music List" to remind them to log in. This message emphasizes that the personal music list is private information and can only be accessed once the user has verified their identity through logging in. **(Figure 3.1)**

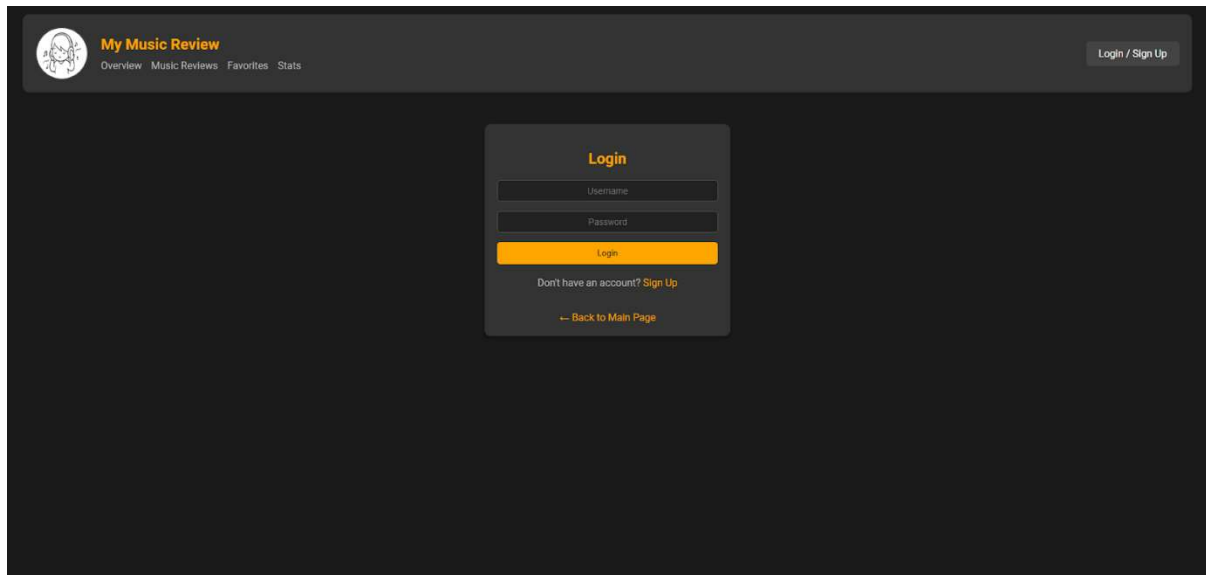


Figure 3.2 Login Interface

After clicking on the "Log In" button displayed on the main screen, the website will show a few fields for users to log in, such as username and password. **(Figure 3.2)**

If the user already has an account, they only need to enter their registered username and password, then click the "Login" button to access the website.

The website also has a login verification system. If the login is correct, the user will be granted access to their account. If the login is incorrect, error messages will appear to help the user quickly adjust their input.

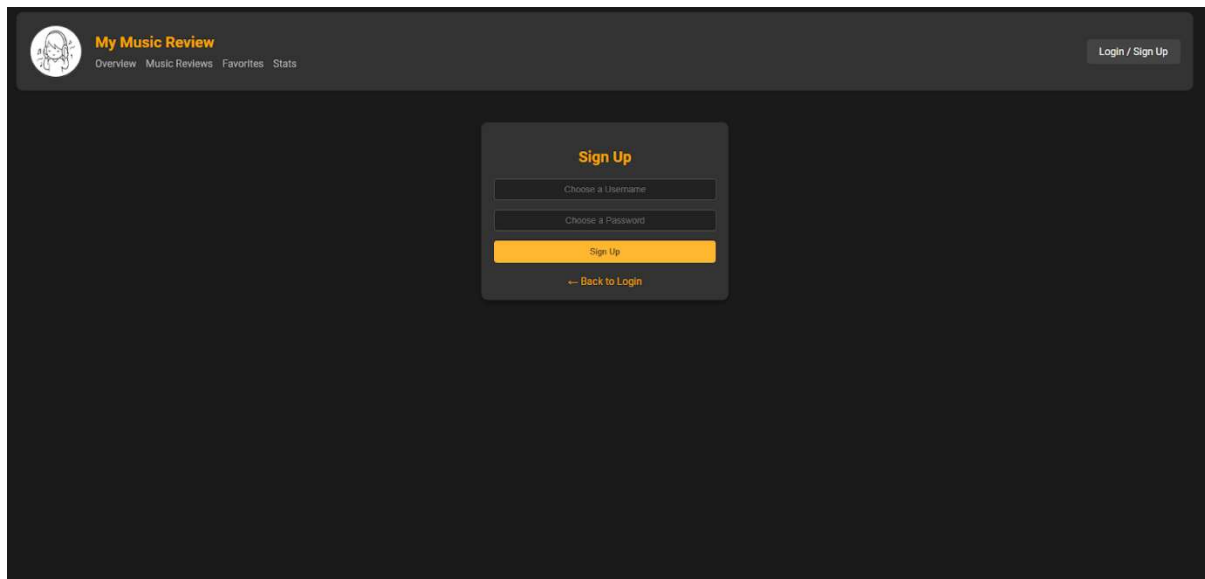


Figure 3.3 Sign Up Interface

In case the user does not have an account, the interface will display a registration section for them. The "Choose a Username" field allows the user to enter a name they want to use, while the "Choose a Password" field is for creating a password for their account. After entering the necessary information, the user should click the "Sign Up" button. Once the account is registered, the user needs to click on "Back to Login" to return to the login page if they already have an account. **(Figure 3.3)**



Figure 3.4 Logout Interface

When wanting to log out of their account, the user simply needs to click the "Logout" button displayed on the screen to end their session. After that, a confirmation message will appear on the main page of the website asking the user to confirm their desire to log out. **(Figure 3.4)**

The login and logout features are not only essential steps for protecting personal information but also an effective way to personalize the user experience. Properly implementing these processes will create a safe and convenient online environment for all users.

When successfully logged into the website, the user will see the main interface, which includes sections for "Overview, Music Reviews, Favourites, Stats" so they can access each section according to their needs. The main page will also display various songs in the "Unreviewed Music" section for users to choose from and review.

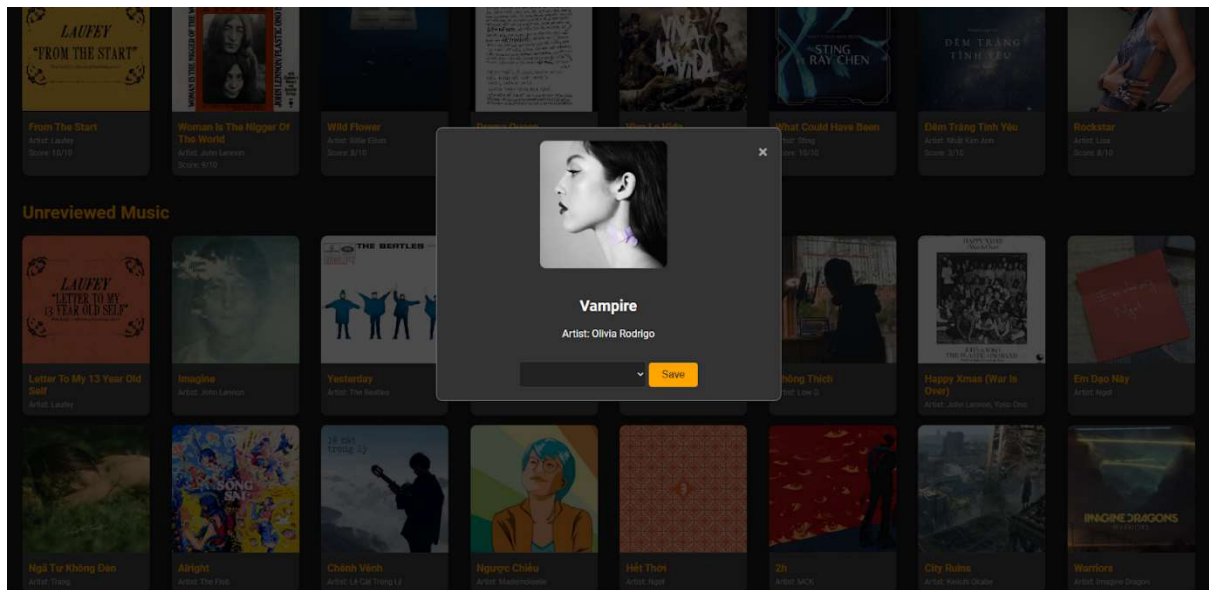


Figure 3.5 Rating of a Song Interface

When users log into the "My Music Review" website, a list of unreviewed songs will appear on the screen. Users have the option to choose any song they want from the "Unreviewed Music" section to rate it on a scale from 1 to 10. This rating will be based on their personal feelings about the music. **(Figure 3.5)**

When users want to rate a song, they simply need to click on the "select" box to choose a suitable score based on their subjective evaluation of that song. Each song can be rated on a scale from 1 to 10. A rating of 1 is the lowest score, indicating that the song is not highly regarded by the user, while scores from 5 to 10 represent songs that are favored and appreciated. Once the user selects an appropriate score, they just need to click the "save" button, and the song will reflect the desired rating. This rating process allows users to express their feelings and subjective opinions about each song, creating a personalized space for their listening experience. **(Figure 3.5)**

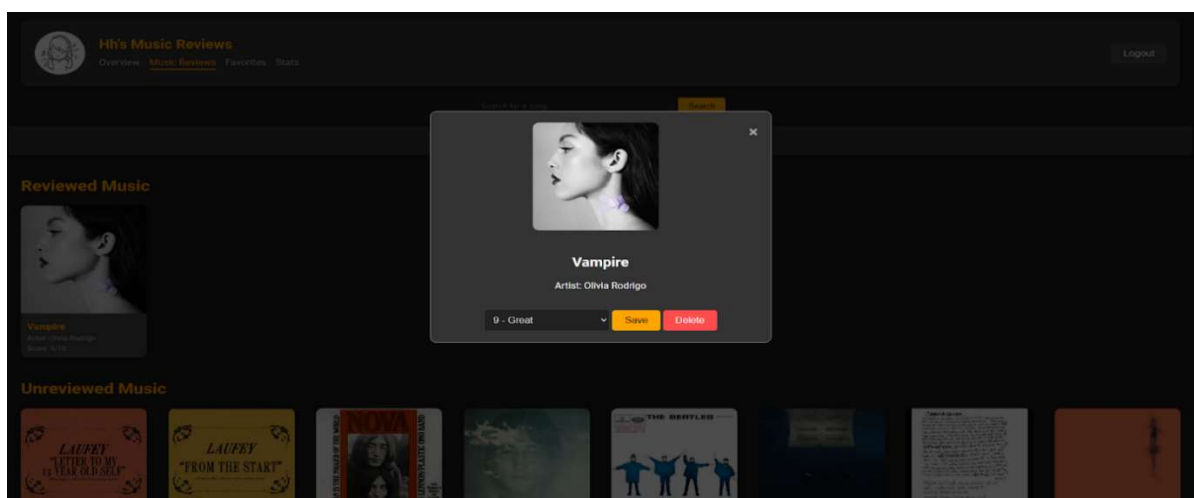


Figure 3.6 Rating of a Song Interface

After users complete the rating of the songs they desire, those songs will appear in the "Reviewed Music" section. This section helps users categorize the songs that have been rated and those that have not, making it easier to track their evaluations. This creates organization and order in their music list. **(Figure 3.6)**

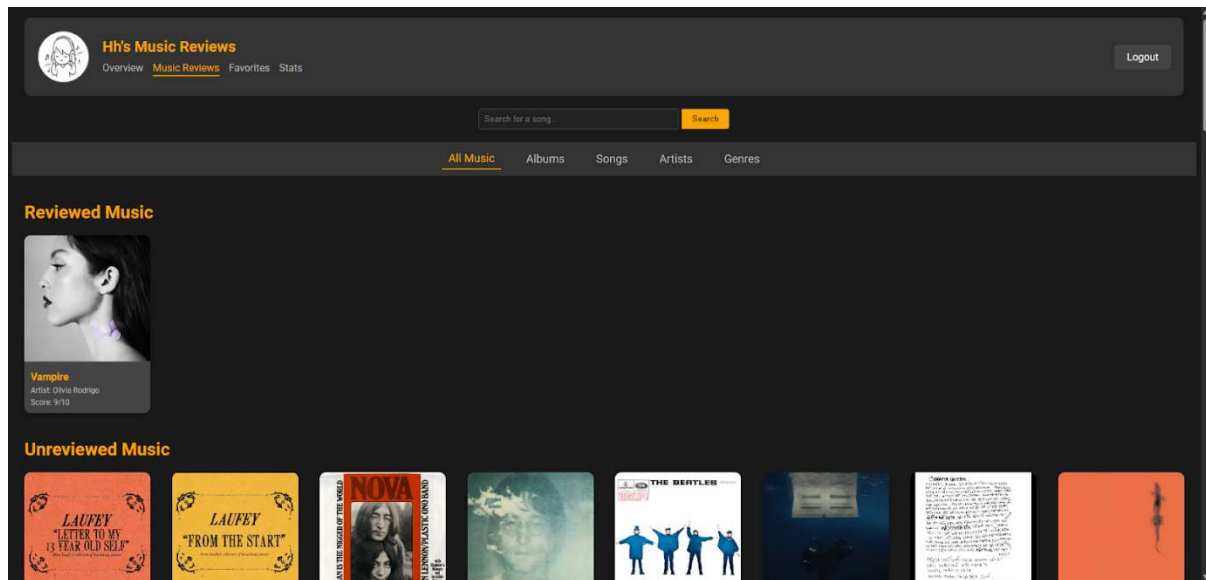


Figure 3.7 Personal Reviewed Music List

The website also has a feature that allows users to change their ratings. After clicking the Delete button, a confirmation message will appear, indicating that the song has been successfully removed. If the song has not been rated again by the user, it will appear in the "Unreviewed Music" section. If the user rates it again, the song will show up in the "Reviewed Music" section with the new score assigned by the user. **(Figure 3.7)**

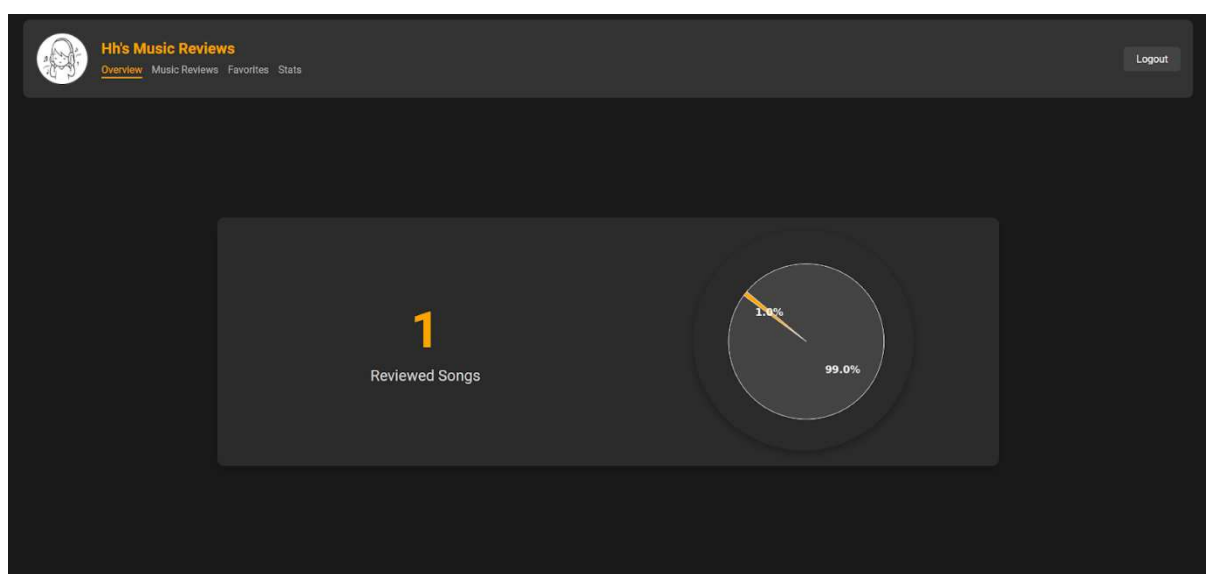


Figure 3.8 Personal Overview Stats Interface

When users click on the Overview section, a summary interface of the songs will appear in percentage form. The yellow portion of the pie chart will represent the percentage of songs that have been rated by the user, while the remaining portion will indicate the percentage of songs that have not been rated. The Overview section will provide users with a general view of their rating activity. **(Figure 3.8)**

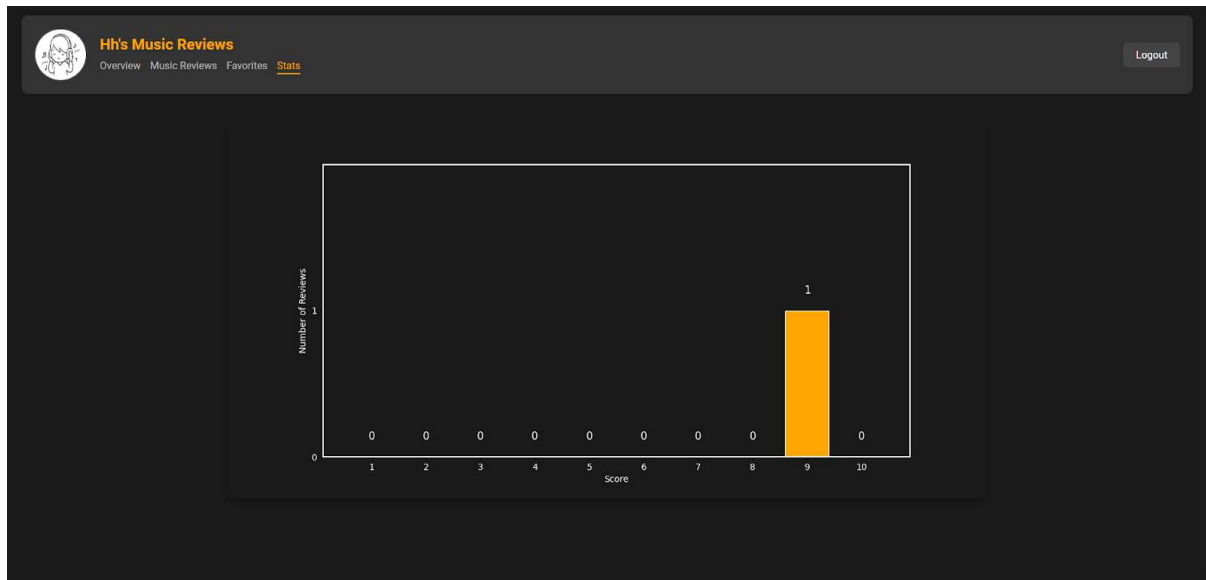


Figure 3.9 Personal Stats Interface

The Stats section helps analyze data to gain a better understanding of the user's preferences and trends. This table will show scores from 1 to 10, indicating how many songs the user has rated at each score. From this, users can see which score has the most rated songs and, conversely, which score has the fewest rated songs. **(Figure 3.9)**

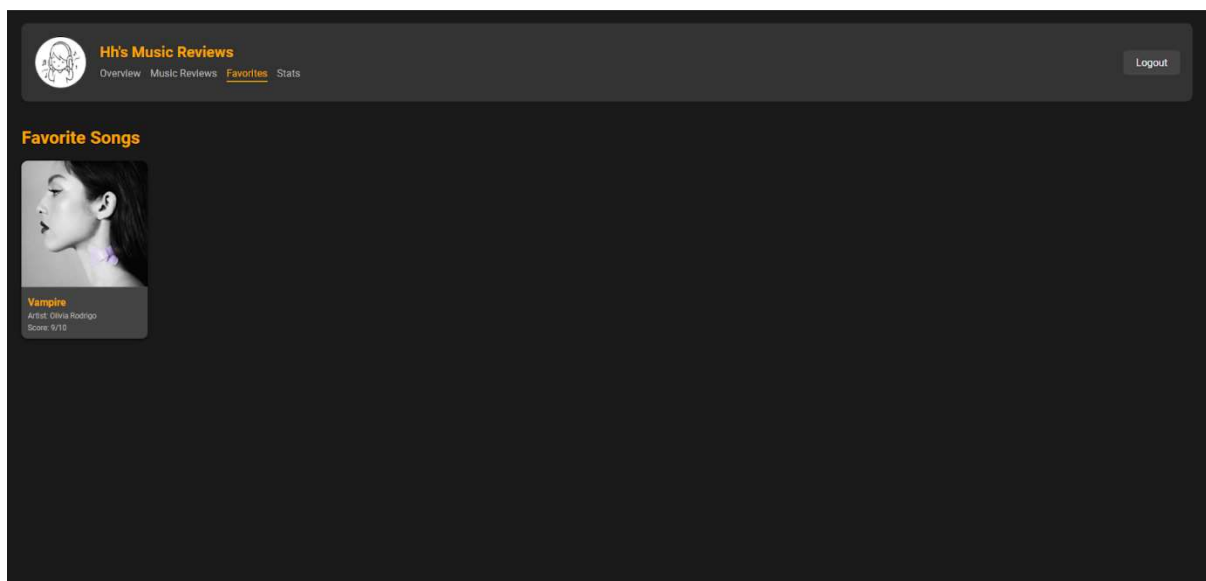


Figure 3.10 Personal Favourite Song Interface

In the Favorite Song section, only the songs with the highest ratings from the user will be displayed. This section provides a dedicated space for the top-rated songs, making it easy for users to quickly find and access their favorite tracks. **(Figure 3.10)**

7. Testing and Evaluation

Testing Method:

Running on Localhost

Localhost is a server environment that runs on a personal computer, allowing for the development and testing of web applications without an internet connection. This creates a safe and efficient space to experiment with the features and functions of the website.

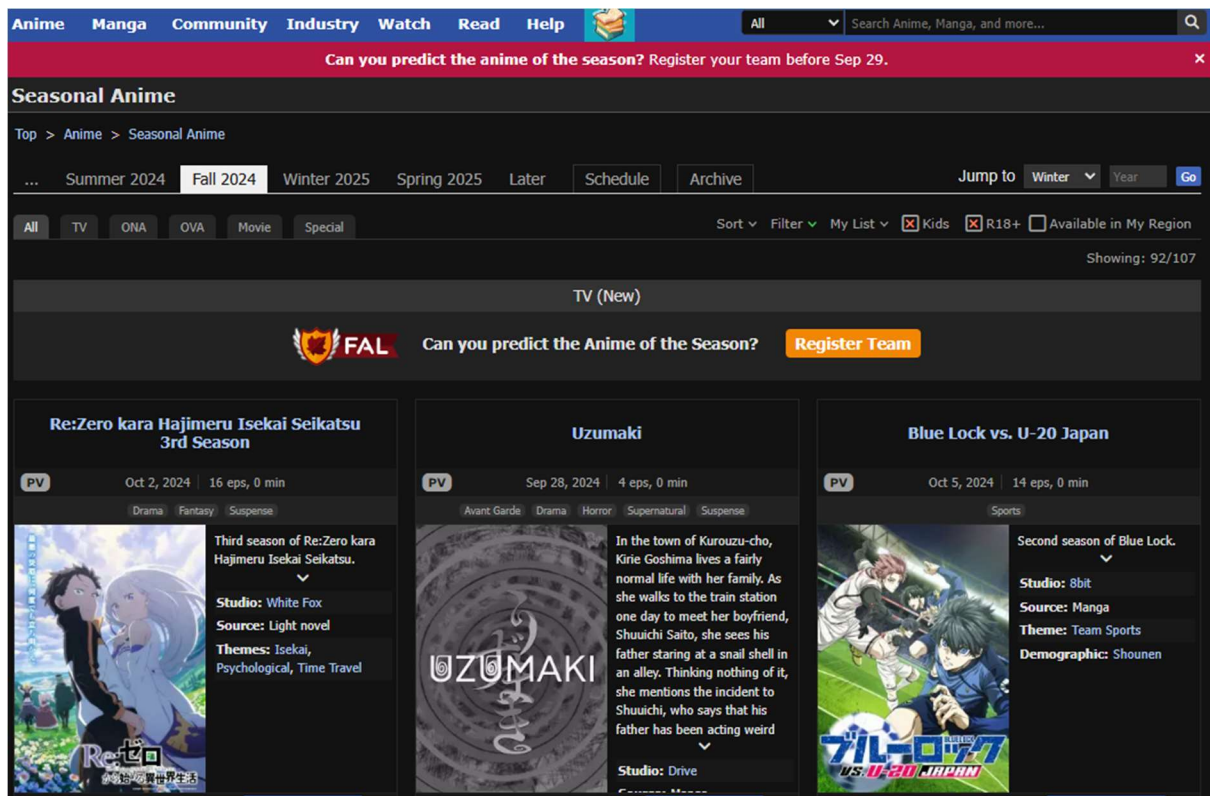
Localhost enables us to test the websites we have created. Running the website on localhost makes data processing quick and does not take as much time as connecting online through hosting. When running on localhost, all requests and responses occur rapidly without being affected by network speed, and there is no need to wait for a connection to a remote server. This is extremely useful when testing complex operations.

Running on localhost helps check the functionalities of the website, such as logging in and out, rating, and more. Testing these functions ensures that users do not encounter issues and that data is stored accurately. Additionally, we can test new features, fix bugs, and refine the interface. Running on localhost is suitable and convenient for learning and practical experimentation. It creates an environment to build and modify a website before deploying it in a real-world setting.

8. Future Feature

8.1 Running Server

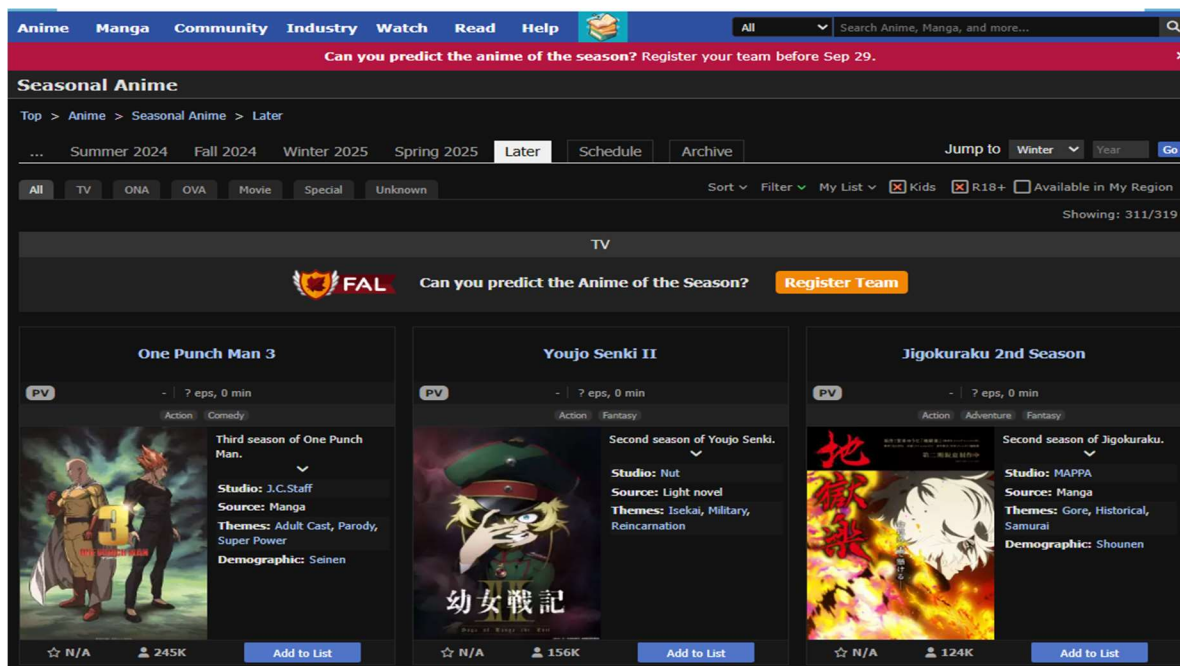
We're excited to roll out our own dedicated website, where everything will be carefully curated by the admin. The idea is to have a clean, organized platform where users can focus on rating, reviewing, and discovering music. The admin will be responsible for uploading all the songs, which means users won't have the ability to upload their own music. This might sound limiting, but it's actually a way to keep the site focused and high-quality. By preventing random uploads, we can ensure that the content stays relevant, organized, and easy to navigate, avoiding the chaos that often comes with user-generated content.



(Source: MAL)

Figure 4.1 Example of Seasonal Song Server Interface

To make things even more engaging, we're going to structure the server by seasons. For example, when Winter 2024 rolls around, we'll feature a collection of songs, albums, and upcoming releases from major artists that fit the vibe of the season. This seasonal approach keeps things fresh and gives users something new to check out each time the season changes. Whether it's the latest big-name release or an indie artist's hidden gem, you'll always have something exciting to look forward to when browsing the site.

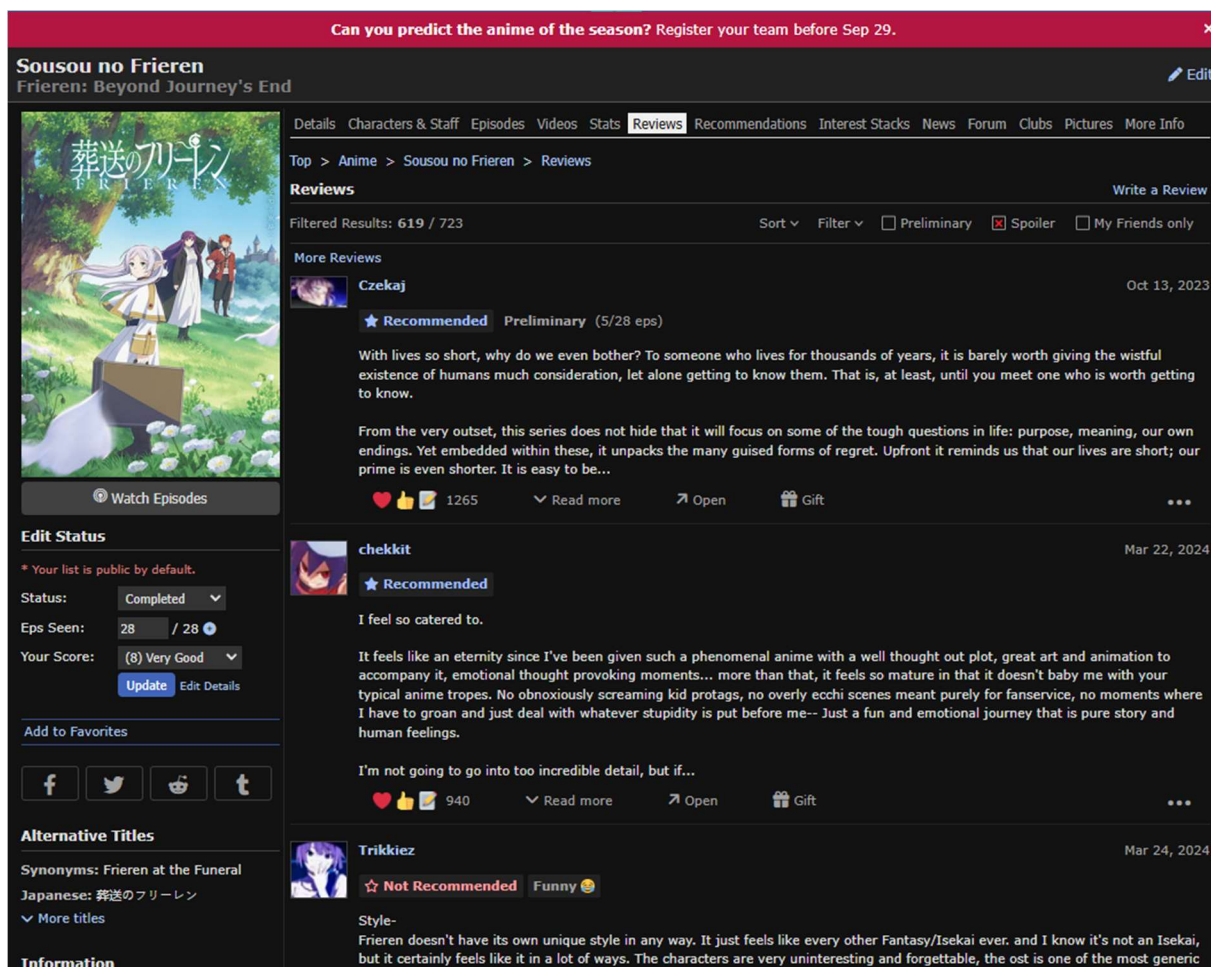


(Source: MAL)

Figure 4.2 Example of Upcoming Song Server Interface

Another key feature we're adding is a **forum** where users can come together to comment, review, and discuss the music they're listening to. It's going to be more than just a place to rate songs; it'll be a community where people can share their thoughts, opinions, and recommendations. If you've ever wanted to chat with other fans about why a certain track hits just right or why an album is a must-listen, the forum will be the perfect space for that. Plus, it adds a layer of social interaction, turning the platform into a lively hub where music lovers can connect and engage with each other.

In short, we're building a platform that not only lets you rate and review music but also connects you with a community of fellow fans. With a clean, admin-curated content system, a fresh seasonal structure, and interactive forums, our goal is to create a space that's fun, engaging, and all about music discovery and discussion.



(Source: MAL)

Figure 4.3 Example of Community Forum Server Interface

9. Conclusion

To wrap things up, My Music List (MML) is shaping up to be a game-changer for music lovers who want more than just a playlist and shuffle button. While popular services like Spotify and Apple Music are great for streaming, they don't really let you dive deep into your personal music preferences or share your thoughts on individual songs. That's where MML steps in—providing a dedicated platform for rating, reviewing, and organizing all your favorite tracks, albums, and artists.

What sets MML apart is its focus on personalization and community. You can rate songs however you like on a scale from 1 to 10, giving you the freedom to reflect your true feelings about the music you listen to. Those ratings aren't just for your eyes, though—MML collects them from everyone and creates a big picture of what the community thinks. It even crunches the numbers and shows how different songs are stacking up in real time, making it easier for you to find the tracks people are loving (or hating).

Beyond that, MML is designed to offer a deeper connection to the music. Its smart features give you detailed stats on your ratings, show you trends across the platform, and even offer recommendations based on your preferences. Plus, you get a secure experience with encrypted passwords and a design that keeps your personal data safe.

On the technical side, MML is built to be simple and efficient. The app's database links user reviews with songs, making it easy for anyone to track their ratings and reviews. The design ensures that as more people join and contribute, the platform stays organized and easy to use, keeping the user experience smooth and the data reliable.

In short, MML is more than just another music app—it's a space for real music fans to express themselves, discover new tracks, and connect with a community of fellow music lovers. It's the perfect blend of personalization, security, and community interaction, offering something unique for anyone who's passionate about music. Whether you're rating your all-time favorite album or hunting for something new to listen to, MML helps make that journey fun, insightful, and rewarding.

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